CST414	DEEP LEARNING	CATEGORY	L	T	P	CREDIT	YEAR OF INTRODUCTION
		PEC	2	1	0	3	2019

Preamble: Deep Learning is the recently emerged branch of machine learning, particularly designed to solve a wide range of problems in Computer Vision and Natural Language Processing. In this course, the building blocks used in deep learning are introduced. Specifically, neural networks, deep neural networks, convolutional neural networks and recurrent neural networks. Learning and optimization strategies such as Gradient Descent, Nesterov Accelerated Gradient Descent, Adam, AdaGrad and RMSProp are also discussed in this course. This course will helps the students to attain sound knowledge of deep architectures used for solving various Vision and NLP tasks. In future, learners can master modern techniques in deep learning such as attention mechanisms, generative models and reinforcement learning.

Prerequisite: Basic understanding of probability theory, linear algebra and machine learning

Course Outcomes: After the completion of the course, the student will be able to

CO1	Illustrate the basic concepts of neural networks and its practical issues (**Cognitive Knowledge Level: Apply)**
CO2	Outline the standard regularization and optimization techniques for deep neural network (**Cognitive Knowledge Level: understand)**
CO3	Implement the foundation layers of CNN (pooling, convolutions) **(Cognitive Knowledge Level: Apply)**
CO4	Implement a sequence model using recurrent neural networks **(Cognitive Knowledge Level: Apply)**
CO5	Use different neural network/deep learning models for practical applications. **(Cognitive Knowledge Level: Apply)**

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Mapping of course outcomes with program outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	✓		✓									✓
CO2	✓	✓	✓		✓							✓
CO3	✓		✓		✓							✓
CO4	✓	✓	✓		✓							✓
CO5	✓	✓	✓	✓	✓							✓

Abstract POs defined by National Board of Accreditation			
PO#	**Broad PO**	**PO#**	**Broad PO**
PO1	Engineering Knowledge	**PO7**	Environment and Sustainability
PO2	Problem Analysis	**PO8**	Ethics
PO3	Design/Development of solutions	**PO9**	Individual and team work
PO4	Conduct investigations of complex problems	**PO10**	Communication
PO5	Modern tool usage	**PO11**	Project Management and Finance
PO6	The Engineer and Society	**PO12**	Life long learning

Assessment Pattern

Bloom's Category	Continuous Assessment Tests		End Semester Examination Marks (%)
	Test 1 (%)	Test 2 (%)	
Remember	30	30	30
Understand	30	30	30
Apply	40	40	40

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Analyze			
Evaluate			
Create			

Mark Distribution

Total Marks	CIE Marks	ESE Marks	ESE Duration
150	**50**	**100**	**3**

Continuous Internal Evaluation Pattern:

Attendance **10 marks**

Continuous Assessment Tests(Average of Internal Tests 1 & 2) **25 marks**

Continuous Assessment Assignment **15 marks**

Internal Examination Pattern

Each of the two internal examinations has to be conducted out of 50 marks. First series test shall be preferably conducted after completing the first half of the syllabus and the second series test shall be preferably conducted after completing remaining part of the syllabus. There will be two parts: Part A and Part B. Part A contains 5 questions (preferably, 2 questions each from the completed modules and 1 question from the partly completed module), having 3 marks for each question adding up to 15 marks for part A. Students should answer all questions from Part A. Part B contains 7 questions (preferably, 3 questions each from the completed modules and 1 question from the partly completed module), each with 7 marks. Out of the 7 questions, a student should answer any 5.

End Semester Examination Pattern:

There will be two parts; Part A and Part B. Part A contains 10 questions with 2 questions from each module, having 3 marks for each question. Students should answer all questions. Part B contains 2 full questions from each module of which student should answer any one. Each question can have maximum 2 sub-divisions and carries 14 marks.

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Syllabus

Module-1 (Neural Networks)

Introduction to neural networks -Single layer perceptrons, Multi Layer Perceptrons (MLPs), Representation Power of MLPs, Activation functions - Sigmoid, Tanh, ReLU, Softmax. , Risk minimization, Loss function, Training MLPs with backpropagation, Practical issues in neural network training - The Problem of Overfitting, Vanishing and exploding gradient problems, Difficulties in convergence, Local and spurious Optima, Computational Challenges. Applications of neural networks.

Module-2 (Deep learning)

Introduction to deep learning, Deep feed forward network, Training deep models, Optimization techniques - Gradient Descent (GD), GD with momentum, Nesterov accelerated GD, Stochastic GD, AdaGrad, RMSProp, Adam. Regularization Techniques - L1 and L2 regularization, Early stopping, Dataset augmentation, Parameter sharing and tying, Injecting noise at input, Ensemble methods, Dropout, Parameter initialization.

Module-3 (Convolutional Neural Network)

Convolutional Neural Networks – convolution operation, motivation, pooling, Convolution and Pooling as an infinitely strong prior, variants of convolution functions, structured outputs, data types, efficient convolution algorithms.

Module- 4 (Recurrent Neural Network)

Recurrent neural networks – Computational graphs, RNN design, encoder – decoder sequence to sequence architectures, deep recurrent networks, recursive neural networks, modern RNNs LSTM and GRU.

Module-5 (Application Areas)

Applications – computer vision, speech recognition, natural language processing, common word embedding: continuous Bag-of-Words, Word2Vec, global vectors for word representation (GloVe). Research Areas – autoencoders, representation learning, boltzmann machines, deep belief networks.

Text Books

1. Goodfellow, I., Bengio,Y., and Courville, A., Deep Learning, MIT Press, 2016.
2. Neural Networks and Deep Learning, Aggarwal, Charu C.
3. Fundamentals of Deep Learning: Designing Next-Generation Machine Intelligence Algorithms (1st. ed.). Nikhil Buduma and Nicholas Locascio. 2017. O'Reilly Media, Inc.

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Reference Books

1. Satish Kumar, Neural Networks: A Classroom Approach, Tata McGraw-Hill Education, 2004.
2. Yegnanarayana, B., Artificial Neural Networks PHI Learning Pvt. Ltd, 2009.
3. Michael Nielsen, Neural Networks and Deep Learning, 2018

Course Level Assessment Questions

Course Outcome1 (CO1):

1. Suppose you have a 3-dimensional input x = (x1, x2, x3) = (2, 2, 1) fully connected to 1 neuron which is in the hidden layer with activation function sigmoid. Calculate the output of the hidden layer neuron.

2. Design a single layer perceptron to compute the **NAND (not-AND)** function. This function receives two binary-valued inputs $\mathbf{x_1}$ and $\mathbf{x_2}$, and returns 0 if both inputs are 1, and returns 1 otherwise.

3. Suppose we have a fully connected, feed-forward network with no hidden layer, and 5 input units connected directly to 3 output units. Briefly explain why adding a hidden layer with 8 linear units does not make the network any more powerful.

4. Briefly explain one thing you would use a validation set for, and why you can't just do it using the test set.

5. Give a method to fight vanishing gradients in fully-connected neural networks. Assume we are using a network with Sigmoid activations trained using SGD.

6. You would like to train a fully-connected neural network with 5 hidden layers, each with 10 hidden units. The input is 20-dimensional and the output is a scalar. What is the total number of trainable parameters in your network?

Course Outcome 2(CO2):

1. Derive a mathematical expression to show L2 regularization as weight decay. Explain how L2 regularization improves the performance of deep feed forward neural networks.

2. In stochastic gradient descent, each pass over the dataset requires the same number of arithmetic operations, whether we use minibatches of size 1 or size 1000. Why can it nevertheless be more computationally efficient to use minibatches of size 1000?

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3. State how to apply early stopping in the context of learning using Gradient Descent. Why is it necessary to use a validation set (instead of simply using the test set) when using early stopping?

4. Suppose that a model does well on the training set, but only achieves an accuracy of 85% on the validation set. You conclude that the model is overfitting, and plan to use L1 or L2 regularization to fix the issue. However, you learn that some of the examples in the data may be incorrectly labeled. Which form of regularisation would you prefer to use and why?

5. Describe one advantage of using Adam optimizer instead of basic gradient descent.

Course Outcome 3(CO3):

1. Draw and explain the architecture of convolutional neural networks.

2. Consider a convolution layer. The input consists of 6 feature maps of size 20×20. The output consists of 8 feature maps, and the filters are of size 5 x 5. The convolution is done with a stride of 2 and zero padding, so the output feature maps are of size 10 x 10.

    a. Determine the number of weights in this convolution layer.

    b. Determine the number of weights if we made this a fully connected layer, but the number of input and output units are kept the same as in the network.

3. Suppose two people A and B have implemented two neural networks for recognizing handwritten digits from 16 x 16 grayscale images. Each network has a single hidden layer, and makes predictions using a softmax output layer with 10 units, one for each digit class.

    a. A's network is a convolutional net. The hidden layer consists of three 16 x 16 convolutional feature maps, each with filters of size 5 x 5, and uses the logistic nonlinearity. All of the hidden units are connected to all of the output units.

    b. B's network is a fully connected network with no weight sharing. The hidden layer consists of 768 logistic units (the same number of units as in A's convolutional layer).

4. Briefly explain one advantage of A's approach and one advantage of B's approach.

5. Why do the layers in a deep architecture need to be non-linear?

6. Give two benefits of using convolutional layers instead of fully connected ones for visual tasks.

7. You have an input volume of 32 x 32 x 3. What are the dimensions of the resulting volume after convolving a 5 x 5 kernel with zero padding, stride of 1, and 2 filters?

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Course Outcome 4(CO4): .

1. Draw and explain the architecture of LSTM.
2. Name at least one benefit of the LSTM model over the bag-of-vectors model.
3. Give one advantage of GloVe over Skipgram/CBOW models.
4. What are two ways practitioners deal with having two different sets of word vectors **U** and **V** at the end of training both Glove and word2vec?
5. If we have a recurrent neural network (RNN), we can view it as a different type of network by "unrolling it through time". Briefly explain what that means.
6. Briefly explain how “unrolling through time” is related to “weight sharing” in convolutional networks.

Course Outcome 5(CO5):

1. Development a deep learning solution for problems in the domain i) natural language processing or ii) Computer vision
2. Illustrate the workings of the RNN with an example of a single sequence defined on a vocabulary of four words.
3. Is an autoencoder for supervised learning or for unsupervised learning? Explain briefly.
4. Sketch the architecture of an autoencoder network.
5. Describe how to train an autoencoder network.

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Model Question Paper

QP CODE:

Reg No: _____

Name: _____ **PAGES : 4**

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EIGHTH SEMESTER B.TECH DEGREE EXAMINATION, MONTH & YEAR

Course Code: CST414

Course Name: Deep Learning

Max. Marks : 100 **Duration: 3 Hours**

PART A

Answer All Questions. Each Question Carries 3 Marks

1. Discuss the limitation of a single layer perceptron with an example.
2. List the advantages and disadvantages of sigmoid and ReLU activation functions.
3. Derive weight updating rule in gradient descent when the error function is a) mean squared error b) cross entropy.
4. Discuss methods to prevent overfitting in neural networks.
5. What happens if the stride of the convolutional layer increases? What can be the maximum stride? Explain.
6. Draw the architecture of a simple CNN and write short notes on each block.
7. How does a recursive neural network work?
8. List down the differences between LSTM and RNN.
9. Illustrate the use of deep learning concepts in Speech Recognition.

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10. What is an autoencoder? Give one application of an autoencoder

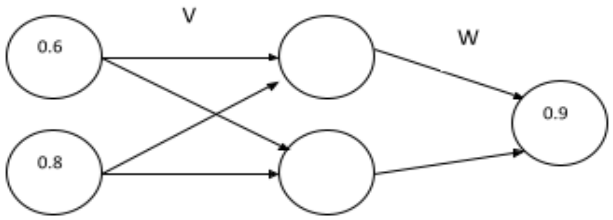
(10x3=30)

Part B

(Answer any one question from each module. Each question carries 14 Marks)

11. (a) Update the parameters in the given MLP using gradient descent with learning rate as 0.5 and activation function as ReLU. Initial weights are given as **(10)**

$V = 0.1\ 0.2 \quad 0.1\ 0.1 \quad W = 0.1\ 0.1$

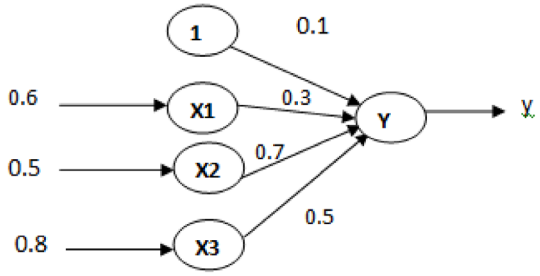


(b) Explain the importance of choosing the right step size in neural networks. **(4)**

OR

12. (a) Draw the architecture of a multi-layer perceptron. Derive update rules for parameters in the multi-layer neural network through the gradient descent **(10)**

(b) Calculate the output of the following neuron Y if the activation function is a **(4)**



binary sigmoid.

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13. (a) Explain, what might happen in ADAGRAD, where momentum is expressed as $\Delta w_t = -\eta g_t / \sqrt{(\sum_{\tau=1}^{t} g_\tau^2)}$ where the denominator computes the L2 norm of all previous gradients on a per-dimension basis and □ is a global learning rate shared by all dimensions. **(6)**

(b) Differentiate gradient descent with and without momentum. Give equations for weight updation in GD with and without momentum. Illustrate plateaus, saddle points and slowly varying gradient. **(8)**

OR

14. (a) Suppose a supervised learning problem is given to model a deep feed forward neural network. Suggest solutions for the following a) small sized dataset for training b) dataset with unlabeled data c) large data set but data from different distribution. **(9)**

(b) Describe the effect in bias and variance when a neural network is modified with more number of hidden units followed with dropout regularization **(5)**

15. (a) Draw and explain the architecture of Convolutional Neural Networks **(8)**

(b) Suppose that a CNN was trained to classify images into different categories. It performed well on a validation set that was taken from the same source as the training set but not on a testing set, which comes from another distribution. What could be the problem with the training of such a CNN? How will you ascertain the problem? How can those problems be solved? **(6)**

OR

16. (a) What is the motivation behind convolution neural networks? **(4)**

(b) Discuss all the variants of the basic convolution function. **(10)**

17. (a) Describe how an LSTM takes care of the vanishing gradient problem. Use some hypothetical numbers for input and output signals to explain the concept. **(8)**

(b) Draw and explain the architecture of Recurrent Neural Networks **(6)**

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18. (a) Explain the application of LSTM in Natural Language Processing. **(8)**

(b) Discuss the architecture of GRU. **(6)**

19. (a) Explain any two word embedding techniques **(8)**

(b) Explain the merits and demerits of using Autoencoders in Computer Vision. **(6)**

OR

20. (a) Illustrate the use of representation learning in object classification. **(7)**

(b) Compare Boltzmann Machine with Deep Belief Network. (7)

Teaching Plan

No	Contents	No. of Lecture Hours (36 hrs)
	Module-1 (Neural Networks) (7 hours)	
1.1	Introduction to neural networks -Single layer perceptrons	1
1.2	Multi Layer Perceptrons (MLPs), Representation Power of MLPs	1
1.3	Activation functions - Sigmoid, Tanh, ReLU, Softmax. , Risk minimization, Loss function	1
1.4	Training MLPs with backpropagation	1
1.5	Illustration of back propagation algorithm	1
1.6	Practical issues in neural network training - The Problem of Overfitting, Vanishing and exploding gradient problems	1
1.7	Difficulties in convergence, Local and spurious Optima, Computational Challenges.	1
	Module-2 (Deep learning) (9 hours)	
2.1	Introduction to deep learning, Deep feed forward network	1
2.2	Training deep models, Concepts of Regularization and optimization,	1

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2.3	Gradient Descent (GD), GD with momentum,	1
2.4	Nesterov accelerated GD, Stochastic GD,	1
2.5	AdaGrad, RMSProp, Adam,	1
2.6	L1 and L2 regularization, Early stopping, Dataset augmentation,	1
2.7	Parameter sharing and tying, Injecting noise at input, Ensemble methods	1
2.8	Parameter sharing and tying, Injecting noise at input, Ensemble methods	1
2.9	Dropout, Parameter initialization.	
Module-3 (Convolutional Neural Network) (6 hours)		
3.1	Convolutional Neural Networks – convolution operation	1
3.2	motivation, pooling	1
3.3	Convolution and Pooling as an infinitely strong prior	1
3.4	Variants of convolution functions	1
3.5	structured outputs, data types.	1
3.6	Efficient convolution algorithms.	1
Module- 4 (Recurrent Neural Network) (5 hours)		
4.1	Recurrent neural networks – Computational graphs, RNN design	1
4.2	Encoder – decoder sequence to sequence architectures	1
4.3	Deep recurrent networks, recursive neural networks	1
4.4	Modern RNNs LSTM	1
4.5	GRU	1
Module-5 (Application Areas)(9 hours)		
5.1	Computer vision. (TB1: Section 12.2)	1
5.2	Speech recognition. (TB1: Section 12.3)	1
5.3	Natural language processing. (TB1: Section 12.4)	1
5.4	Common Word Embedding - Continuous Bag-of-Words, Word2Vec (TB3: Section 2.6)	1

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5.5	Common Word Embedding - Global Vectors for Word Representation(GloVe) (TB3: Section 2.9.1- Pennington 2014)	1
5.6	Brief introduction on current research areas - Autoencoders, Representation learning. (TB3: Section 4.10)	1
5.7	Brief introduction on current research areas - representation learning. (TB3: Section 9.3)	1
5.8	Brief introduction on current research areas - Boltzmann Machines, Deep belief networks. (TB1: Section 20.1, TB3 Section 6.3)	1
5.9	Brief introduction on current research areas - Deep belief networks. (TB1: Section 20.3)	1

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